

Eventual admissible equal-sided sizes at factorial order

Proof with completed independent mathematical reviews

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This note combines the short signed fair target, the polynomial-growth Ore bound, and the new order-doubling positive-prefix completion. It improves the earlier eventual-size threshold, without changing the known asymptotic upper bound for the *smallest* fair size.

Theorem 1. *There is an absolute constant C such that, for every $n \geq 2$, every integer M satisfying*

$$\prod_{p \leq n} p \mid M, \quad M \geq \exp(Cn \log(n+1))$$

is the common number of faces of a permutation-fair collection of n equiprobable finite dice with globally distinct face labels. More precisely, there are two attainable common counts $A, A+P$ with

$$P = \prod_{p \leq n} p, \quad P \mid A, \quad A+P \leq \exp(Cn \log(n+1)).$$

Every positive multiple of P at least $A(A-P)/P$ is then attainable.

Proof. Write $T_a = \exp(aH)$ in the reduced integral-signature group, with $H = X_1 + \cdots + X_n$. The reviewed constructive signed-signature theorem and its base-digit compression, in `quantitative_eventual_sizes.tex`, give a signed word for T_P of length at most

$$L_* = 2^{2n+3} n^4 n! P = \exp(O(n \log(n+1))).$$

This formula is the short-signed-target lemma of that note; the older completion and conductor estimates there are not used here.

The reviewed growth/Ore lemma in `linear_logarithmic_ore_bound.tex` says that a signed word of length L is a right fraction uv^{-1} , with both positive lengths at most

$$\lceil 16D_n L \log(8D_n L) \rceil, \quad D_n = \sum_{j=1}^n j \frac{n!}{(n-j)!} \leq enn!.$$

Consequently $T_P = uv^{-1}$ with $|u|, |v| \leq H_n$ for some $H_n = \exp(O(n \log(n+1)))$.

Apply `order_doubling_prefix_completion.tex` to v . There is a positive suffix c with $vc = T_A$ and

$$A \leq (H_n + 1) \exp(O(n \log(n+1))) = \exp(O(n \log(n+1))).$$

Here equations of words denote equations of signatures. Then $T_P T_A = uv^{-1}vc = uc$ is positive, so both A and $A+P$ are attainable common counts. The subset-factorial divisibility condition forces $P \mid A$: restrict to any p dice, for each prime $p \leq n$, and use $p! \mid A^p$.

Put $a = A/P$. Concatenating balanced fair words adds their common counts, because $T_x T_y = T_{x+y}$. Thus every nonnegative integer combination of a and $a+1$, multiplied by P , is attainable. Every integer $t \geq a(a-1)$ is such a combination: write $t = qa + r$ with $0 \leq r < a$, so $q \geq a-1 \geq r$ and $t = (q-r)a + r(a+1)$. Therefore every admissible count at least $Pa(a-1) = A(A-P)/P$ is attainable. Its bound is still $\exp(O(n \log(n+1)))$, proving the theorem. Finally a positive word is realized by assigning successive globally distinct integers to its positions and interpreting equal letters as faces of the corresponding equiprobable die. $\square$

Scope and attribution. The theorem says that above a factorial-order threshold the elementary prime divisibility conditions are sufficient for *every* equal-sided size. It does not distinguish $\exp(\Theta(n))$ from $\exp(\Theta(n \log n))$ for the minimum total size, and does not prove polynomial equalization of an arbitrary optimal family. The order-doubling input uses the published Kuperberg–Lovett–Peled t -wise permutation theorem. All other inputs are identified by their separate proof notes; no novelty claim about the classical tools is made.